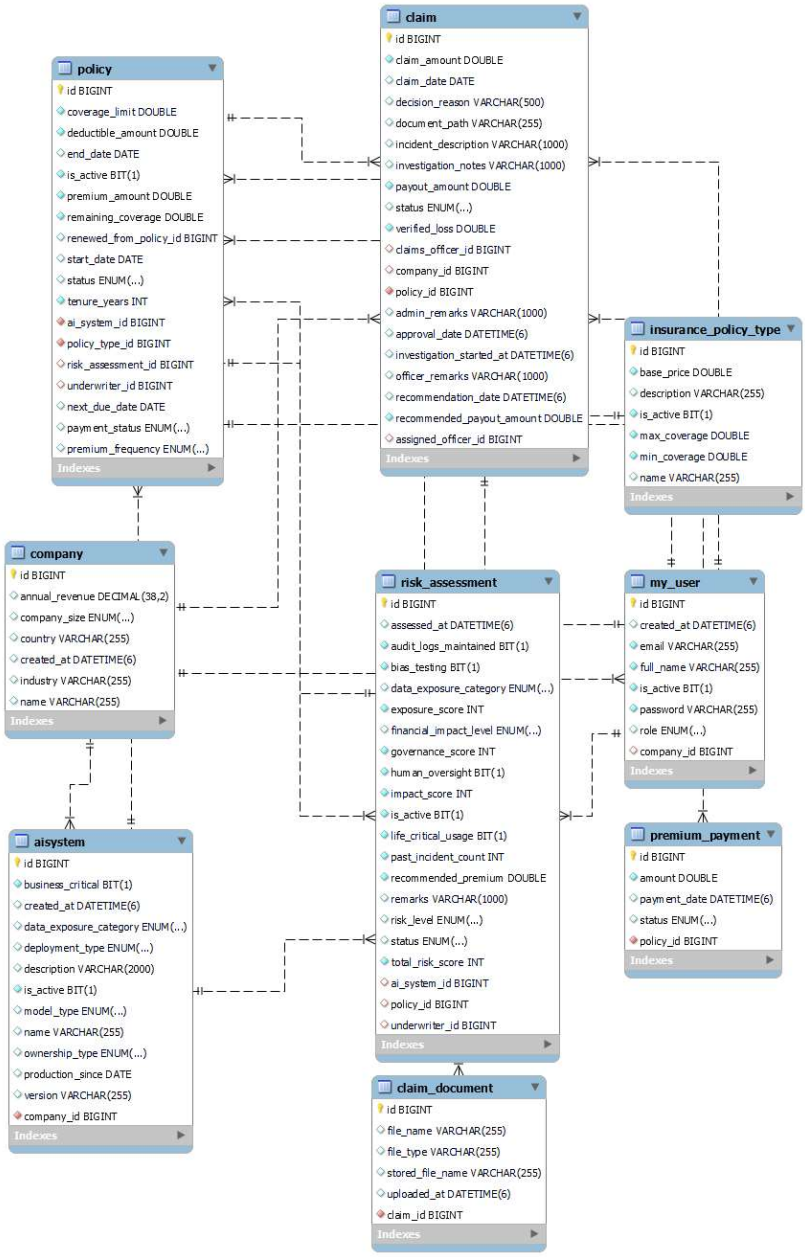
Structural UML Diagrams

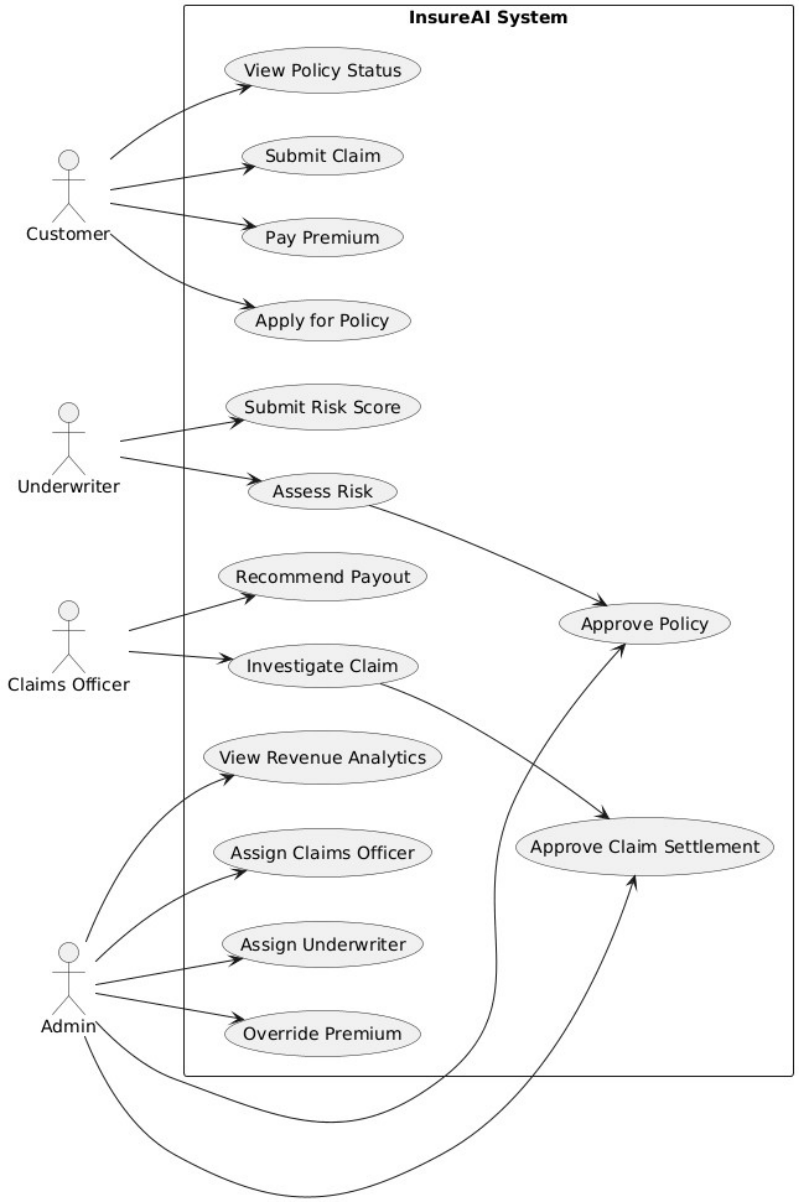
ER Diagram

The Entity Relationship (ER) Diagram represents the database structure of the InsureAI platform. It illustrates the major entities such as User, Company, AI System, Policy, Risk Assessment, Claim, Premium Payment, and Claim Document along with their relationships. The diagram helps in understanding how data is organized and how different modules interact through relational constraints. It ensures data integrity and supports efficient transaction management.



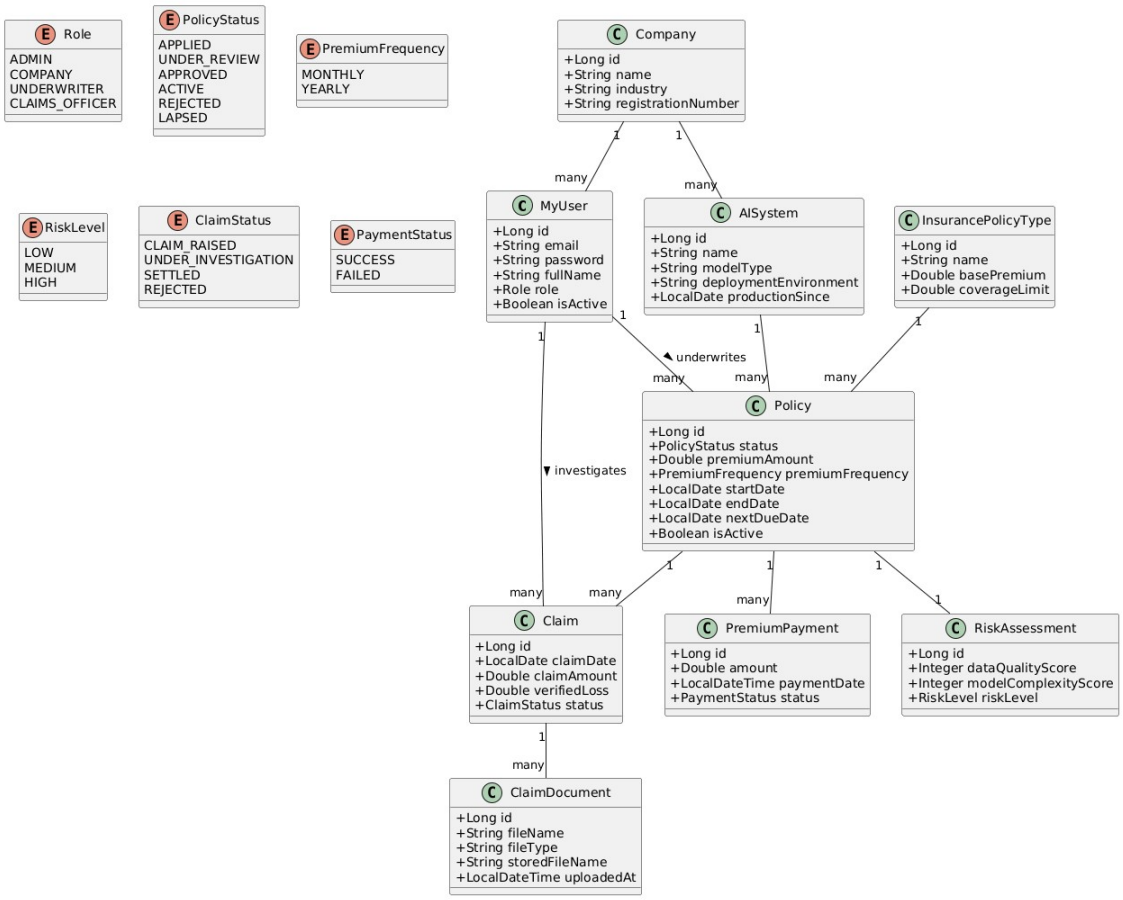
Use Case Diagram

The Use Case Diagram depicts the interaction between system users and the core functionalities of the InsureAI platform. It highlights how different actors such as Customer, Underwriter, Claims Officer, and Admin perform operations like policy application, risk assessment, premium payment, claim processing, and revenue monitoring. This diagram provides a high-level view of system behaviour from the user's perspective.



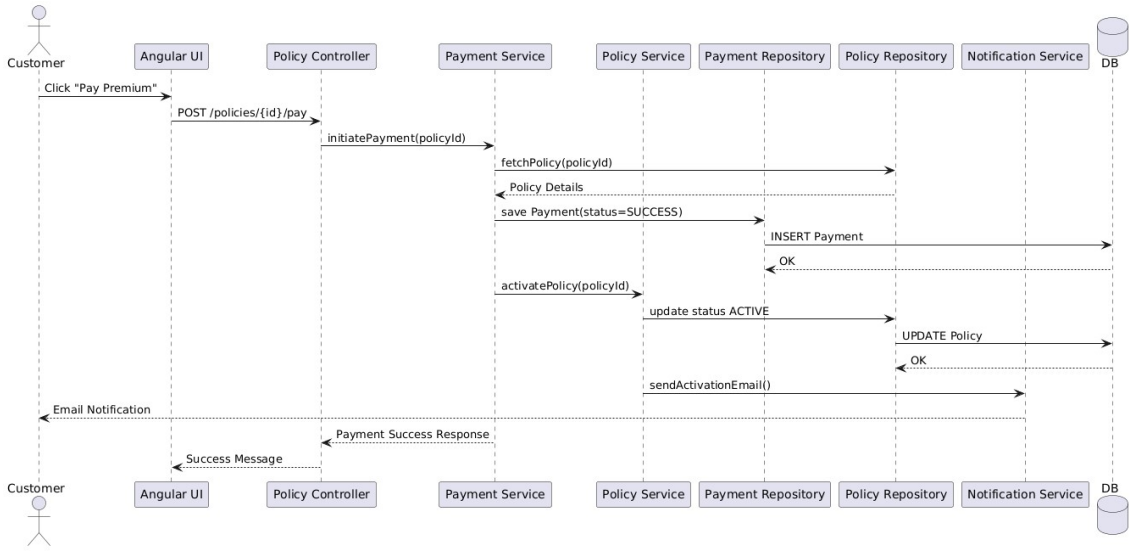
Class Diagram

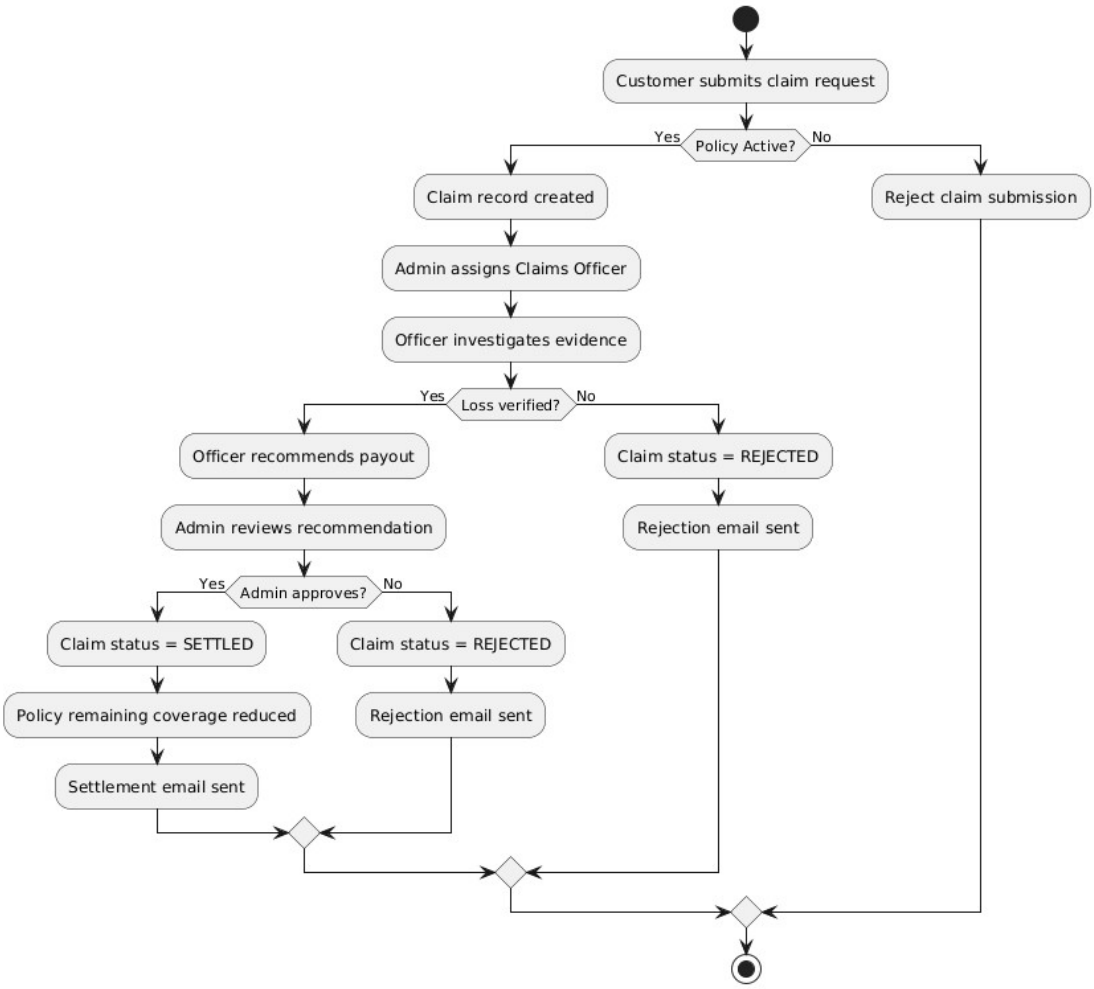
The Class Diagram illustrates the static structural design of the InsureAI system. It shows key domain classes such as Policy, Claim, User, Risk Assessment, Premium Payment, and Company along with their attributes and relationships. This diagram helps developers understand object interactions and supports implementation of the backend data model using object-oriented principles.



Sequence Diagram

The Sequence Diagram represents the step-by-step interaction flow between system components during critical operations such as premium payment and policy activation. It demonstrates how the user interface, controller layer, service layer, database, and notification services communicate in a time-ordered sequence. This diagram helps in understanding runtime behaviour and system response flow.



Activity Diagram

The Activity Diagram describes the procedural workflow of business processes such as claim submission and settlement. It highlights decision points, parallel activities, and sequential execution steps involved in verifying claims and approving payouts. This diagram provides clarity on operational logic and helps stakeholders visualize process automation within the system.